

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

Annexure A

Debugging & Optimisation Task

Code of Conduct:

Shanika (192511387)

I Betty A. (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Betty

20/20
R. Sub

Annexure A

Debugging & Optimisation Task

1. A network administrator has configured a workstation with the following settings:

- **IP Address:** 192.168.10.65
- **Subnet Mask:** 255.255.255.192 (/26)
- **Default Gateway:** 192.168.10.1

However, the workstation is unable to communicate with other systems configured in the subnet **192.168.10.0/26**.

Task

Debug the network configuration by:

- Identifying the subnet to which the configured IP address belongs.
- Determining the network address, broadcast address, and valid host range.
- Verifying whether the configured IP address and gateway are valid.
- Identifying the configuration errors.
- Optimizing the configuration by assigning an appropriate IP address and default gateway.
- Calculating the total number of usable host addresses after correction.

2. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

3. An organization has allocated the network **192.168.1.0/24** into several departments using the subnet masks **/26**, **/27**, and **/28**. Some departments report insufficient IP addresses, while others have a large number of unused addresses.

Task

Analyze and optimize the subnet allocation by:

- Identifying the network address, broadcast address, and host range for each subnet.
- Calculating the number of usable and unused IP addresses.
- Identifying inefficient subnet allocations.
- Debugging the addressing plan by locating subnet allocation mistakes.
- Recommending an optimized subnetting scheme using VLSM to improve address utilization while supporting current departmental requirements.

4. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

5. A router contains the following routing table.

Destination Network	Next Hop
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default Route	10.0.0.1

Users report that packets destined for **192.168.2.100** are not reaching the intended network.

Task

Debug the routing configuration by:

- Identifying the matching routing entry using the longest-prefix match.
- Determining the next-hop router selected for packet forwarding.

- Verifying whether the destination IP belongs to the intended subnet.
- Identifying any routing configuration errors.
- Optimizing the routing table to improve packet forwarding efficiency.
- Identifying the route that will be used if no matching destination exists.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Identification	20%	Accurately identifies all errors and root causes with clear understanding	Identifies most errors with minor gaps	Identifies some errors but misses key issues	Unable to identify errors correctly
Debugging	25%	Applies systematic debugging techniques effectively to resolve issues	Uses appropriate debugging methods with minor errors	Limited debugging approach; requires guidance	Unable to debug or resolve issues
Optimization	20%	Optimizes code by significantly improving time complexity using efficient algorithms	Improves time complexity with minor gaps in efficiency	Limited improvement in time complexity	No improvement; inefficient approach retained
Analysis	20%	Demonstrates strong logical reasoning and clear justification of solutions	Shows reasonable analysis with some justification	Limited analysis; weak reasoning	No analytical reasoning demonstrated
Code Quality	15%	Produces clean, well-structured code with proper comments and documentation	Code is mostly clear with minor documentation gaps	Code lacks structure and proper documentation	Poorly written code with no documentation

Q1. IPv4 Debugging

Given:

IP Address: 192.168.10.65

Subnet Mask: 255.255.255.192 (126)

Default Gateway: 192.168.10.1

a) subnet

192.168.10.65 belongs to the
192.168.10.64/26 subnet.

b) network details

Network Address: 192.168.10.64

Broadcast Address: 192.168.10.127

Valid Host Range: 192.168.10.65 -

192.168.10.126

c) verification:

IP Address (192.168.10.65): valid

Gateway (192.168.10.1): invalid

(belongs to subnet 192.168.10.0/26)

d) configuration Error

The default gateway is in a different subnet

e) optimized configuration

IP Address : 192.168.10.65

Default Gateway : 192.168.10.126

(or any valid host in the same subnet)

f) usable Hosts

$$2^{(32-26)} - 2 = 64 - 2 = 62$$

Q2. IPv6 Debugging

Device A

2001:db8::ff00:42:8329/64

Expanded:

2001:0db8:0000:0000:0000:ff00:0042
:8329

Device B

2001:db8:0:0:1::1/64

Expanded:

2001:0db8:0000:0000:0001:0000:0000:0001

Network Prefix

Device A : 2001:db8:0:0::1/64

Device B : 2001:db8:0:0::1/64

Same subnet : Yes

Addressing issue :

NO prefix mismatch . IF communication

fails, check :

- Incorrect gateway
- Firewall
- Interface status
- Duplicate address

Optimized Address

Eg: Device A : 2001:db8:0:0::10/64

Device B : 2001:db8:0:0::20/64

Available Hosts:

$$2^{64} = 18,446,744,073,709,551,616$$

Q3. Subnet Optimization (VLSM)

126

Network : 192.168.1.0

Broadcast : 192.168.1.63

Host Range : 192.168.1.1 - 192.168.1.62

Usable Hosts : 62

127

Network : 192.168.1.64

Broadcast : 192.168.1.95

Host Range : 192.168.1.65 - 192.168.1.94

Usable Hosts : 30

128

Network : 192.168.1.96

Broadcast : 192.168.1.111

Host Range : 192.168.1.97 - 192.168.1.110

Usable Hosts : 14

Inefficient Allocation

Small departments using 128 waste address

Large departments using 128 run out of addresses.

Optimized VLSM

Large department → 126

Medium department → 127

Small department → 128

This improves address utilization.

Q4. IPv6 Debugging

Device A Expanded:

2001:0db8:0000:0000:0000:ff00:0042:8329

Device B Expand:

2001:0db8:0000:0000:0001:0000:0000:0001

Network prefix: $2001:db8:0:0::/64$

same subnet: yes

suggested addresses:

Device A $\rightarrow 2001:db8:0:0::10/64$

Device B $\rightarrow 2001:db8:0:0::20/64$

Hosts: $2^{64} = 18,446,744,073,709,551,616$

Q5. Routing table debugging

$192.168.1.0/24 - 10.0.0.2$ (Next hop)

$192.168.2.0/24 - 10.0.0.3$

Default - $10.0.0.1$

a) Longest prefix match

Destination IP $192.168.2.100$

Matches:

$192.168.2.0/24$

b) Next Hop

$10.0.0.3$

c) Destination Verification:

192.168.2.100 belongs to 192.168.2.0/24

d) Routing Error

If packets fail:

- wrong next hop
- interface down
- incorrect routing entry

e) Optimization:

Verify next-hop router (10.0.0.3)

Ensure interface is active

Remove incorrect or duplicate routes

f) If no matching route exists

Router uses the default route →

10.0.0.1.

These answers directly address all
the tasks described in your assessment